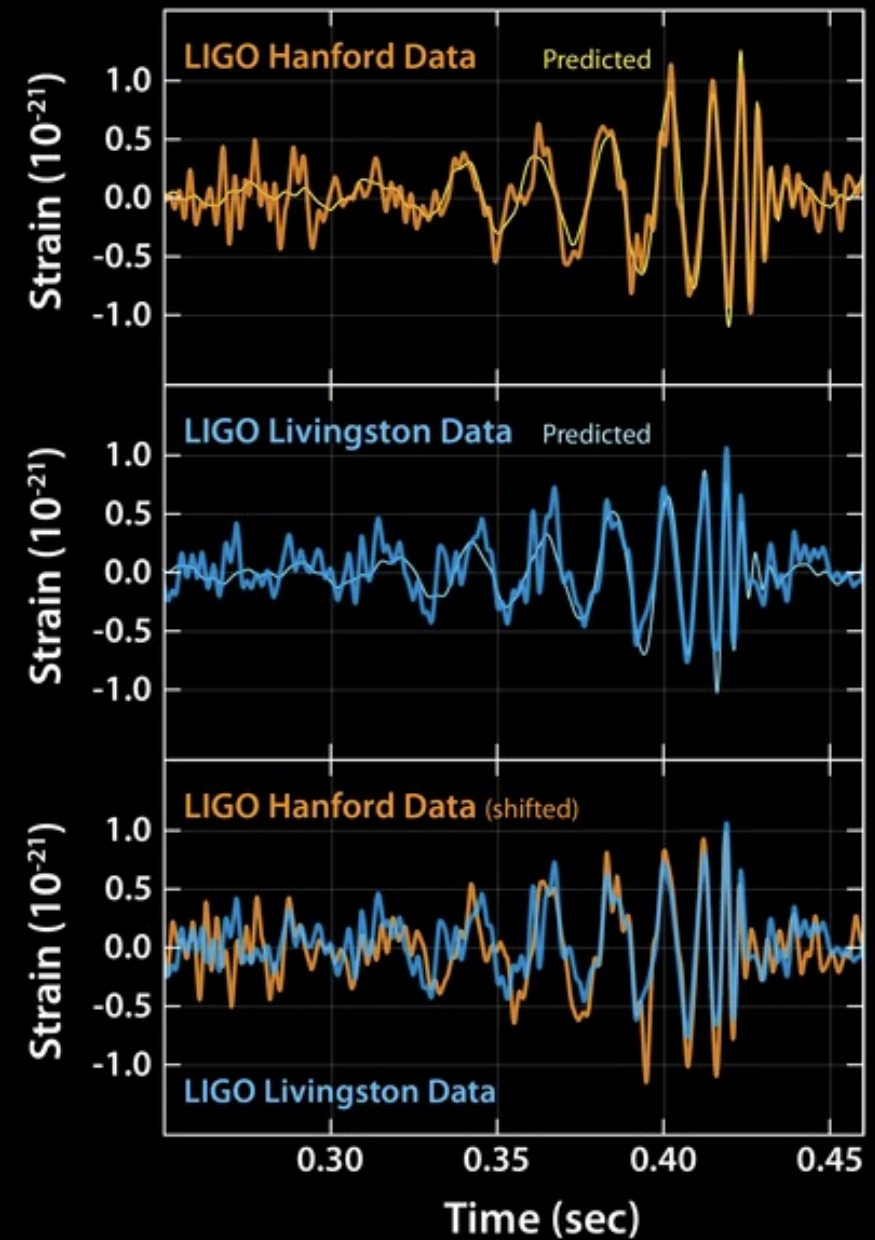
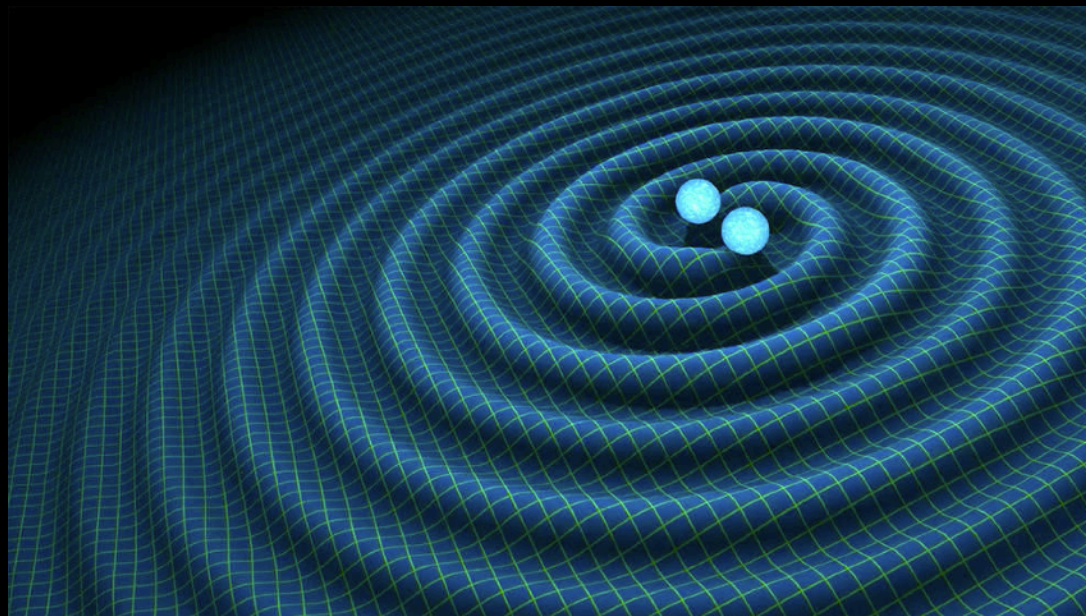
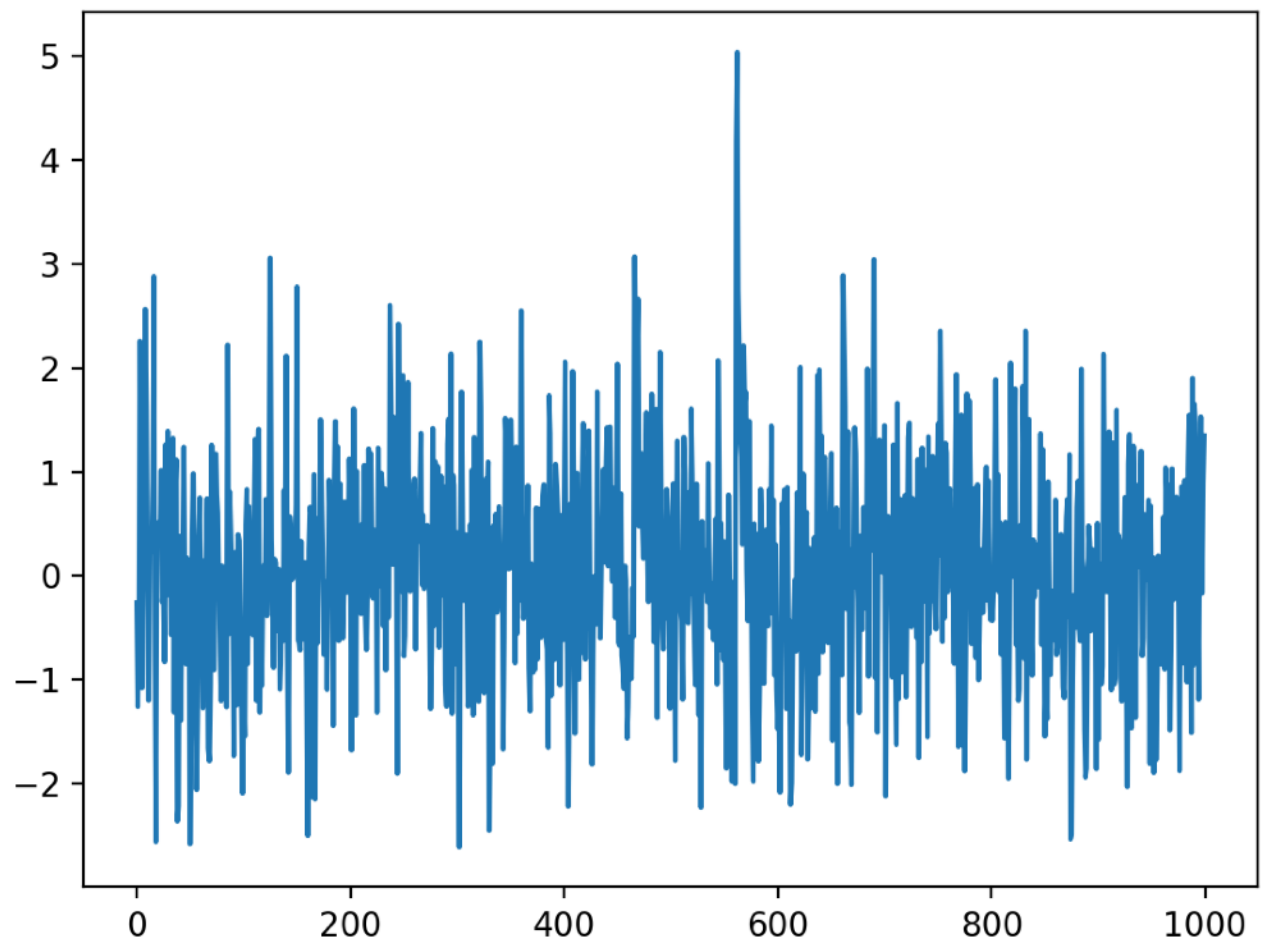


Matched Filters/Ligo



Searching for Signals

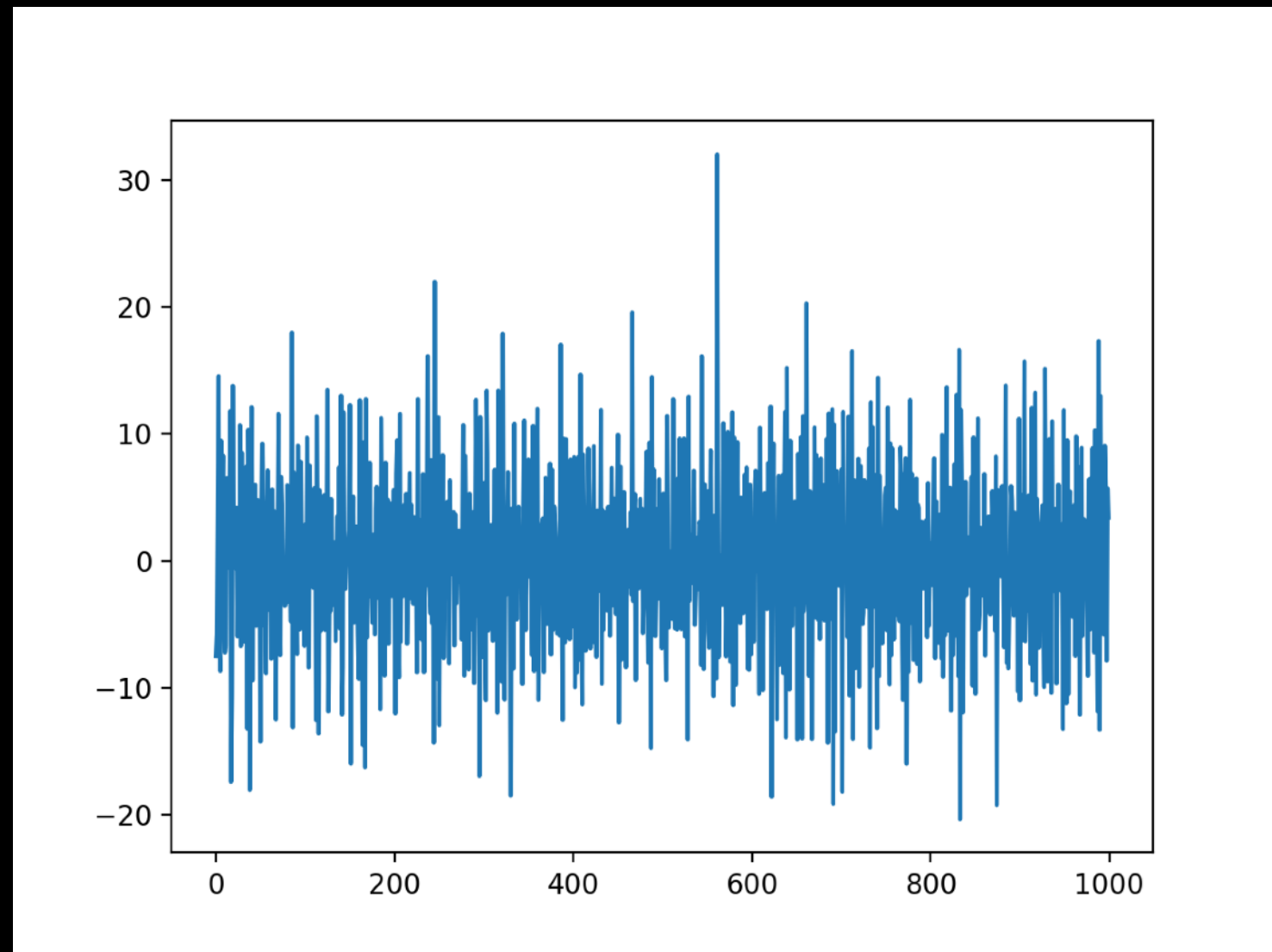
- Simulated data - particle detector.
- Signal decays exponentially when hit by particle.
- Noise is white.
- Where/how energetic were the particles that hit?



Simulated data of exponential-decay detector hit by particles.

Deconvolution

- Simplest attempt is to deconvolve observed data using exponential response.
- How might this work or not work? Why?



Deconvolved version of previous

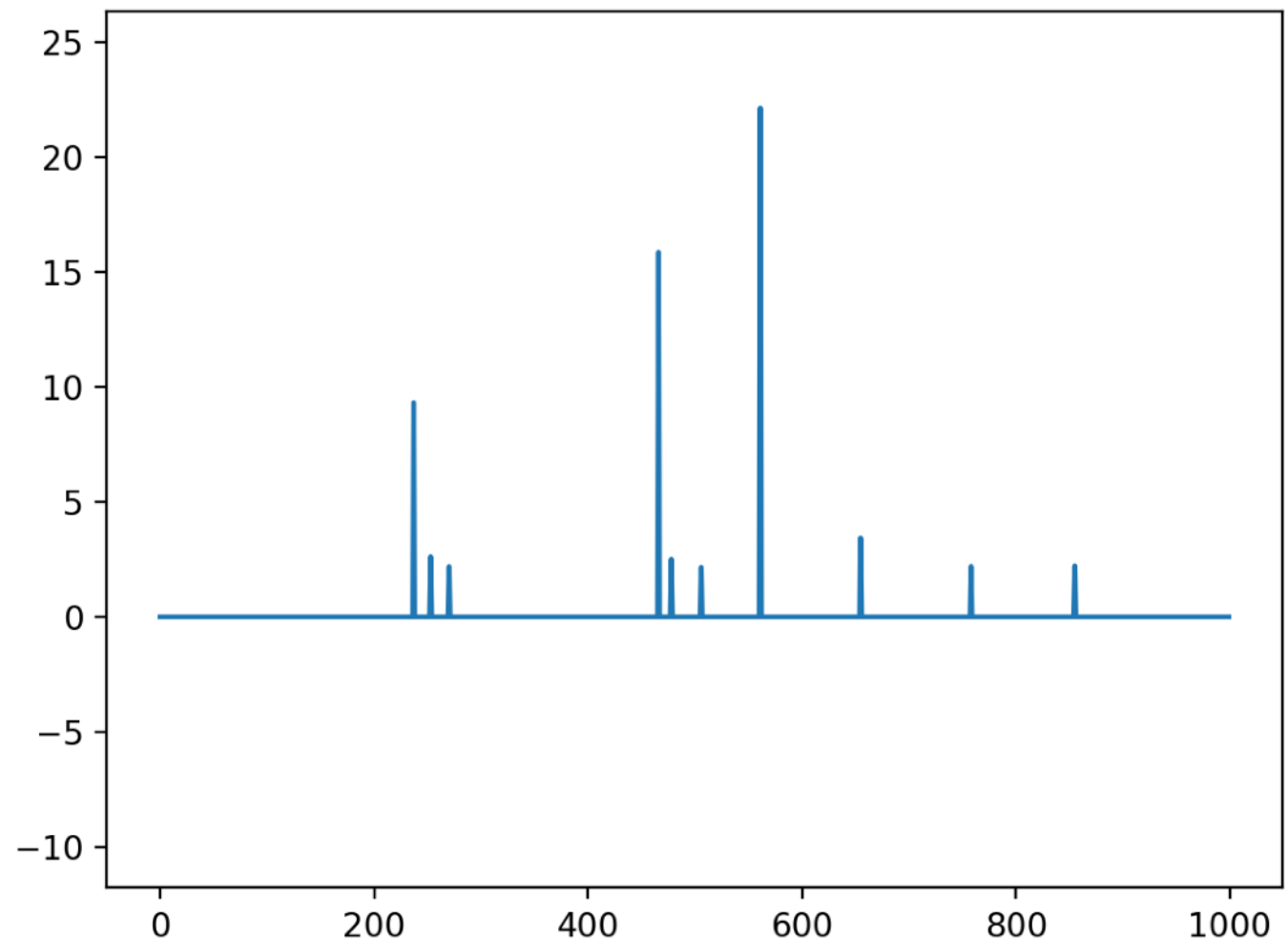
Matched Filter

- We want to search for a signal in data. We don't know where it will be. How do we find it?
- Best fit amplitude for 1-D template A is $A^T N^{-1} d / A^T N^{-1} A$
- We can search many possible locations of template with matched filter, replacing top by correlation of A with $N^{-1} d$ (or $N^{-1} A$ with d) if noise is stationary
- Alternatively, could take correlation of $N^{-1/2} A$ with $N^{-1/2} d$. What would the noise in $N^{-1/2} d$ look like?

NB - relevant operation is cross-correlation, not convolution

MF Output

- Output of MF, vs. true input signal.
- In this case, SNR of MF is twice that of deconvolution.
- MF *multiplies* by noise, deconv. *divides*. MF stays stable in presence of zeros in noise.

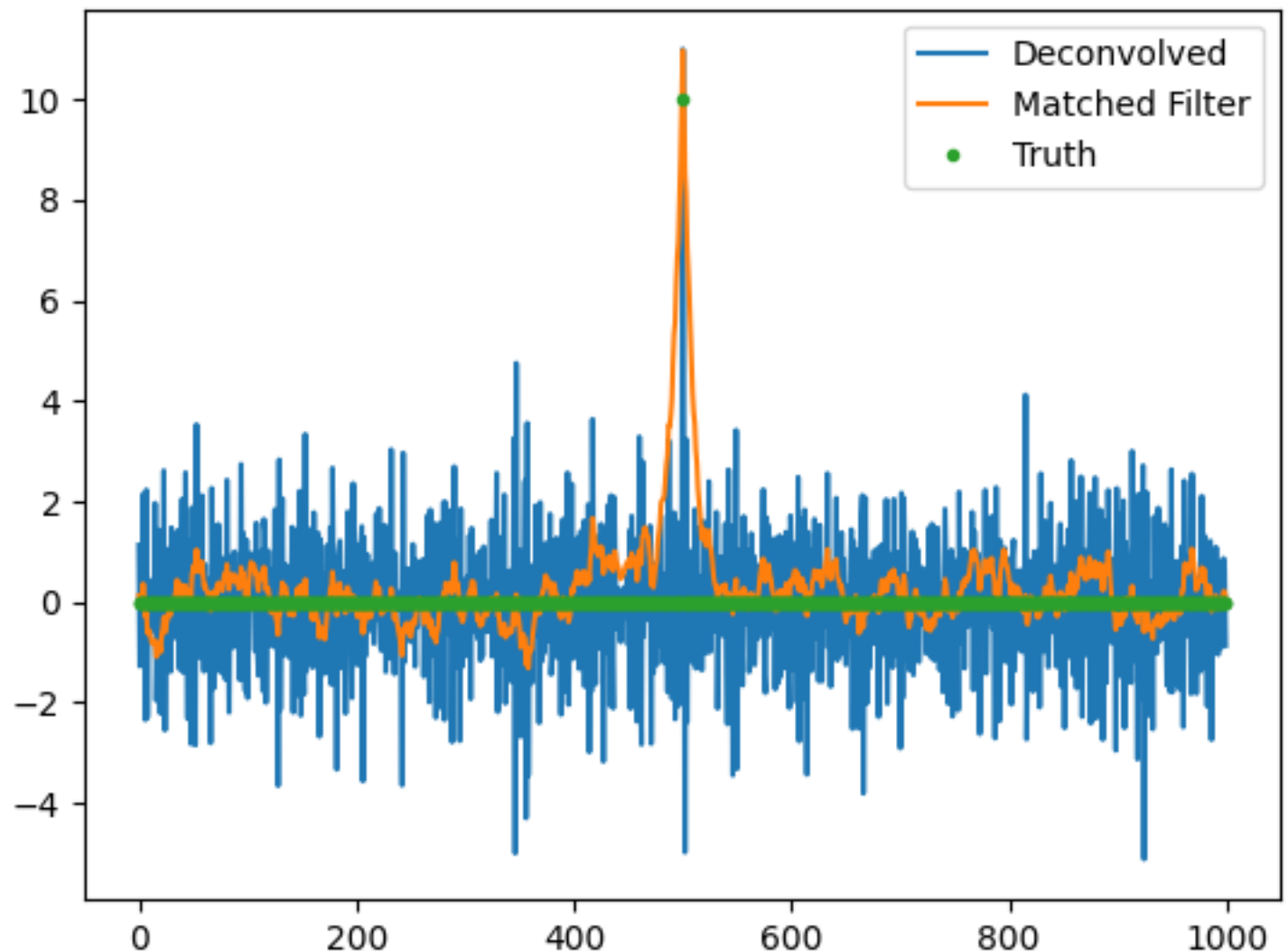


MF vs. Deconv

- Matched filter expects exactly one signal in the data.
- Deconvolution solves for signal at *every* data point.
- If your signals overlap, assumptions for MF are wrong. If they don't, deconv is higher noise.
- Use the right tool for the right job!

MF vs. Deconv Graph

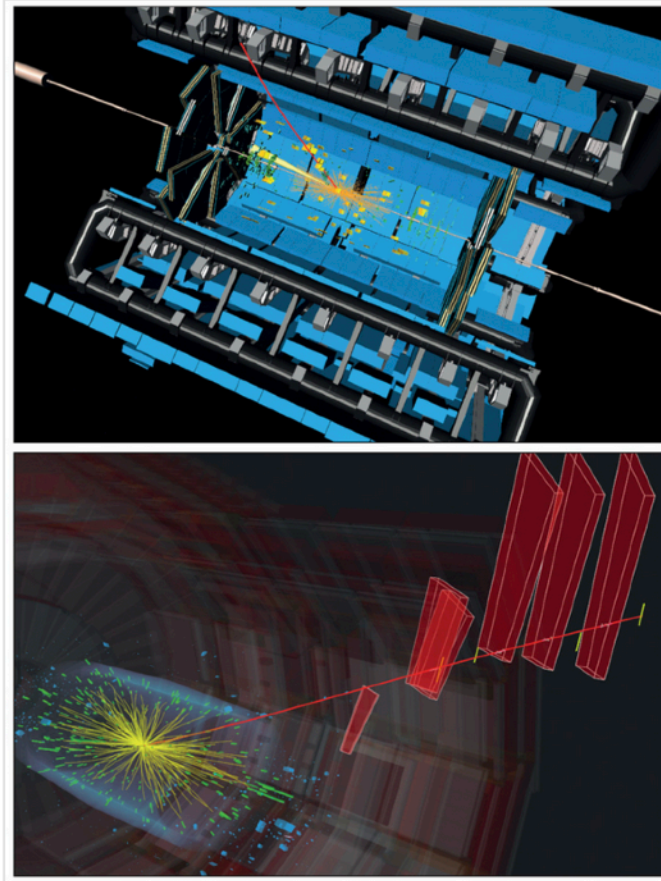
- MF lower noise, but errors more correlated.
- Deconv higher noise, uncorrelated errors.
- Signals closer than MF correlation will interfere.



Output of mf_deconv_snr.py

Next-generation triggers for HL-LHC and beyond

3 May 2024



On your marks ATLAS (top) and CMS (above) events recorded at a collision energy of 13.6 TeV

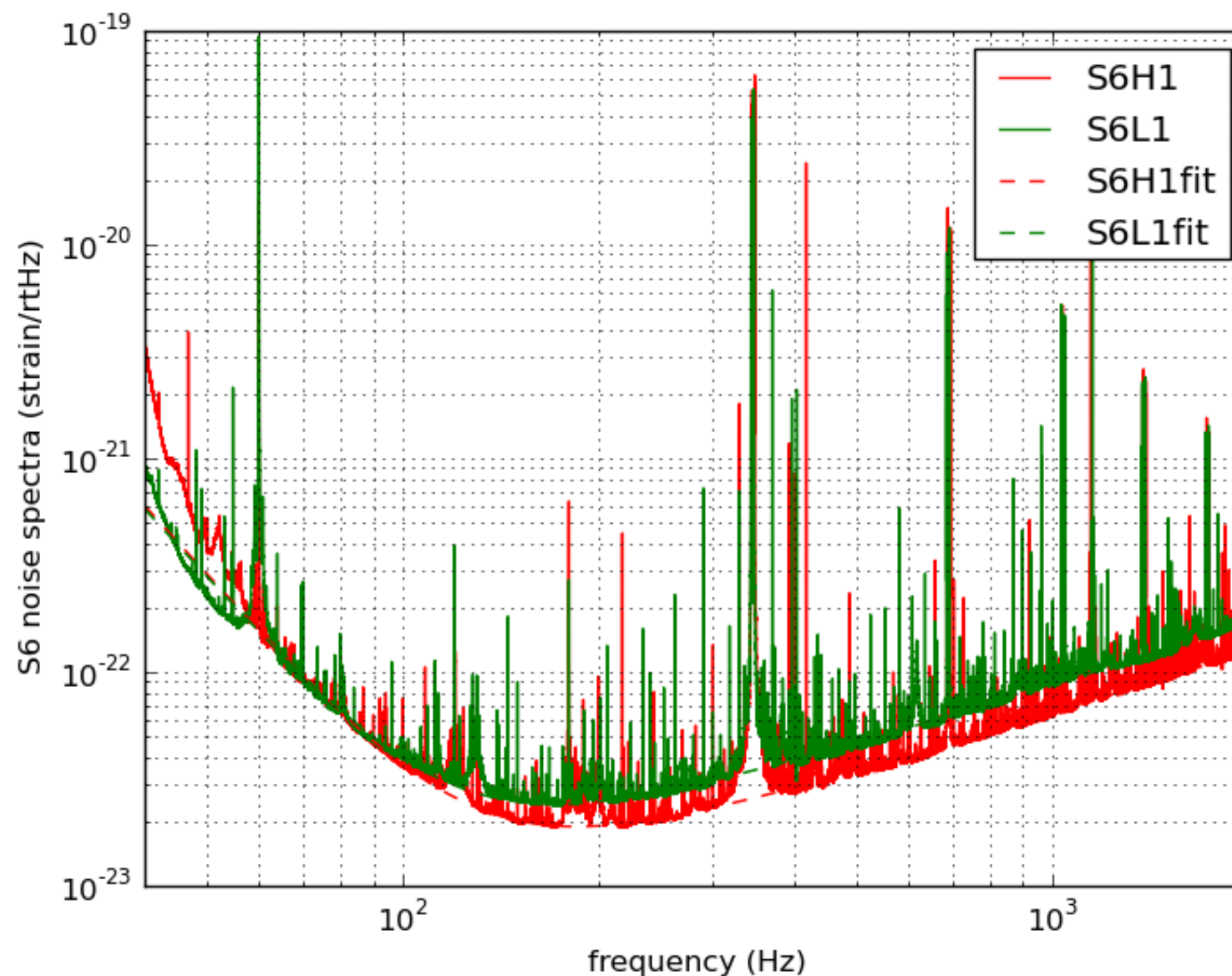
The LHC experiments have surpassed expectations in their ability to squeeze the most out of their large datasets, also demonstrating the wealth of scientific understanding to be gained from improvements to data-acquisition pipelines. Colliding proton bunches at a rate of 40 MHz, the LHC produces a huge quantity of data that must be filtered in real-time to levels that are manageable for offline computing and ensuing physics analysis. When the High-Luminosity LHC (HL-LHC) enters operation from 2029, the data rates and event complexity will further increase significantly.

Following a visit by a group of private donors, in 2023 CERN, in close collaboration with the ATLAS and CMS collaborations, submitted a proposal to the Eric and Wendy Schmidt Fund for Strategic Innovation, which resulted in the award of a \$48 million grant. The donation laid the foundations of the Next Generation Triggers project, which

LIGO Data

- https://github.com/losc-tutorial/LOSC_Event_tutorial
- alternatively, relevant files in ligo directory on course github (but losc-tutorial has more)
- `simple_read_ligo.py` will read for you (once you have h5py installed and working)

First, what should we see for noise?



Power Spectrum Description

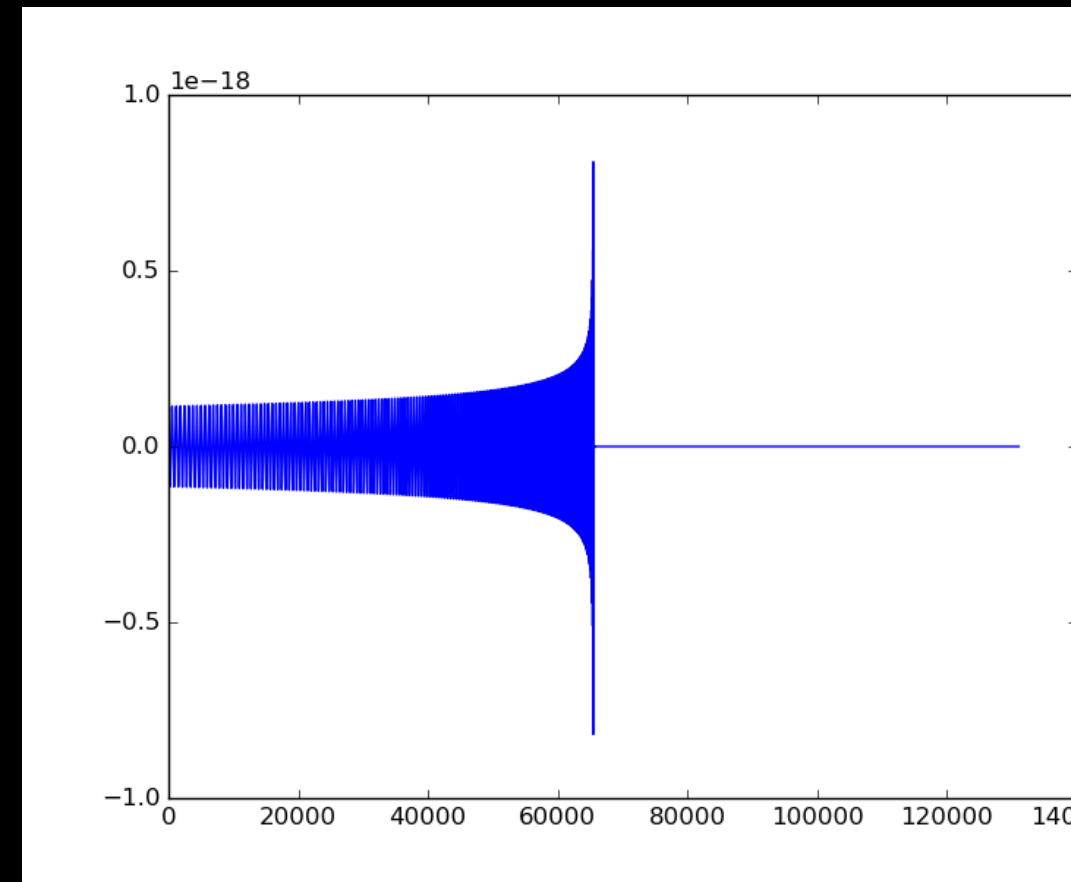
- χ^2 is the same in real or Fourier space
- Modes are uncorrelated in Fourier space. We'll describe noise there.
- SNR^2/mode is set by $(\text{template FT})^2/\text{noise PS}$
- Noise PS is just FT of correlation function

Noise in Fourier Space

- Let F be standard FT/ $\sqrt{n}$. Then $F^{-1} = F^\dagger$.
- $\chi^2 = r^T F^\dagger F N^{-1} F^\dagger F r$ where $r = d - A(m)$.
- Let $\tilde{r} = Fr$ and $\tilde{N} = F N F^\dagger$. We know $\tilde{N}$ is diagonal, so estimate from $\tilde{r}$. If noise $\gg$ signal, then $r \sim d$, so estimate from FT of data.
- Rewrite $\chi^2 = r F^\dagger \tilde{N} F r$. Then we know $N = F^\dagger \tilde{N} F$.
- Matched filter: We need $A^T N^{-1} d$ at every offset, so can correlated A against $N^{-1} d$ or d against $N^{-1} A$. We apply N^{-1} by taking FFT, dividing by (diagonal) $\tilde{N}$, and taking inverse FFT.

Pre-whitening

- One sometimes useful trick is to pre-whiten.
- Template is at right. Where does our SNR come from?
- Can't say because depends on details of N .
- Pre-whitening turns N into I , so can be obvious what's happening.



Pre-whitening

- Re-write least-squares solution: $(N^{-1/2}A)^T(N^{-1/2}A)m=(N^{-1/2}A)^T(N^{-1/2}d)$
- Let $A'=N^{-1/2}A$, $d'=N^{-1/2}d$, and get $A'^TA'm=A'^Td'$. Same as before, but now noise is identity. Hence, pre-whitening.
- $N^{-1/2}$ can be done by dividing by square root of noise.
- Look at pre-whitened template $(N^{-1/2}A)$ to see where you expect useful signal.
- Usual choice for $N^{-1/2}$ is $V\lambda^{-1/2}V^T$.

Fourier Interpretation

- Noise model has same total variance independent of correlation length.
- Looking at FT, long length packs noise power into many long wavelengths. Template has more power on high-frequency scales (good SNR)
- Short length spreads out power over many many modes, dropping average noise power. Template well above noise on large scales (good SNR).
- Intermediate packs all its noise into same scales as template. Never have good SNR.

When your noise looks like your signal,
you're going to have a bad day...

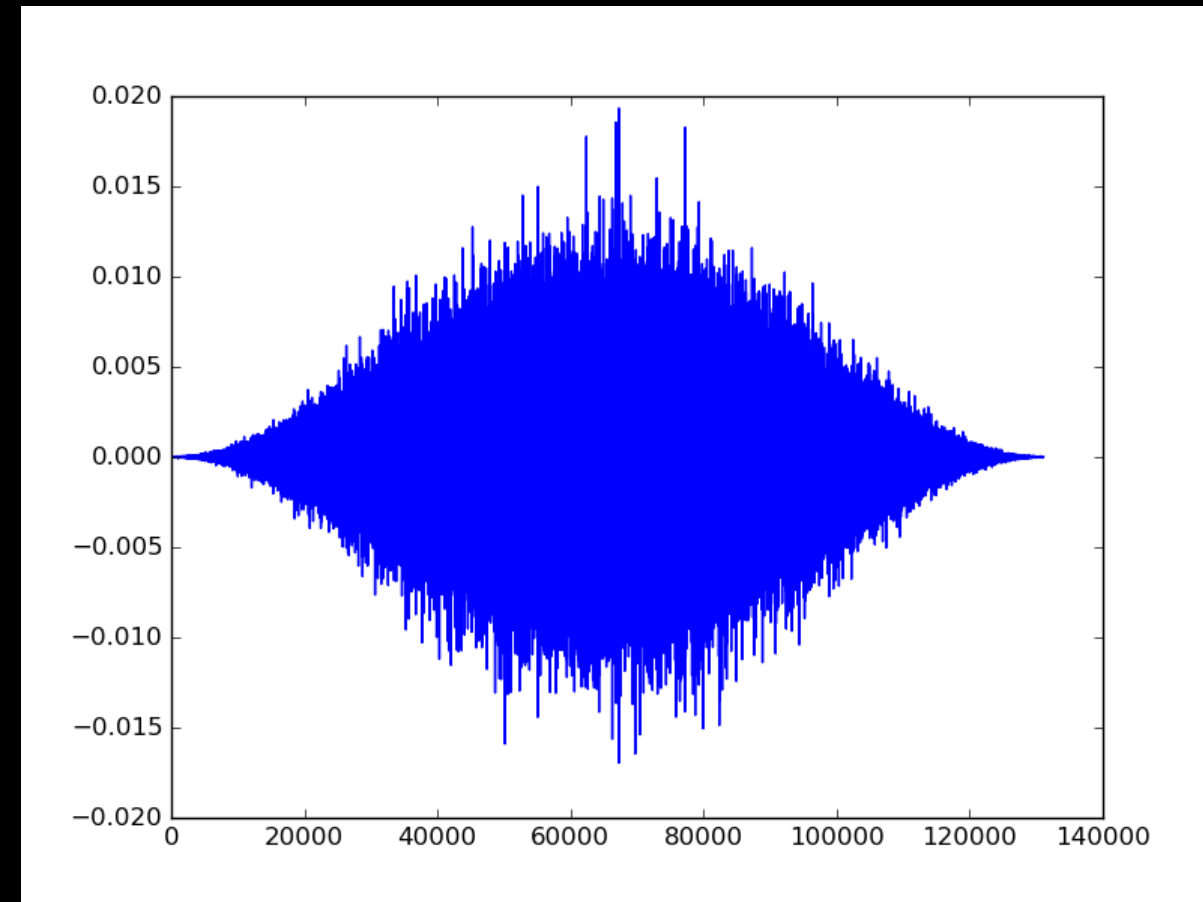
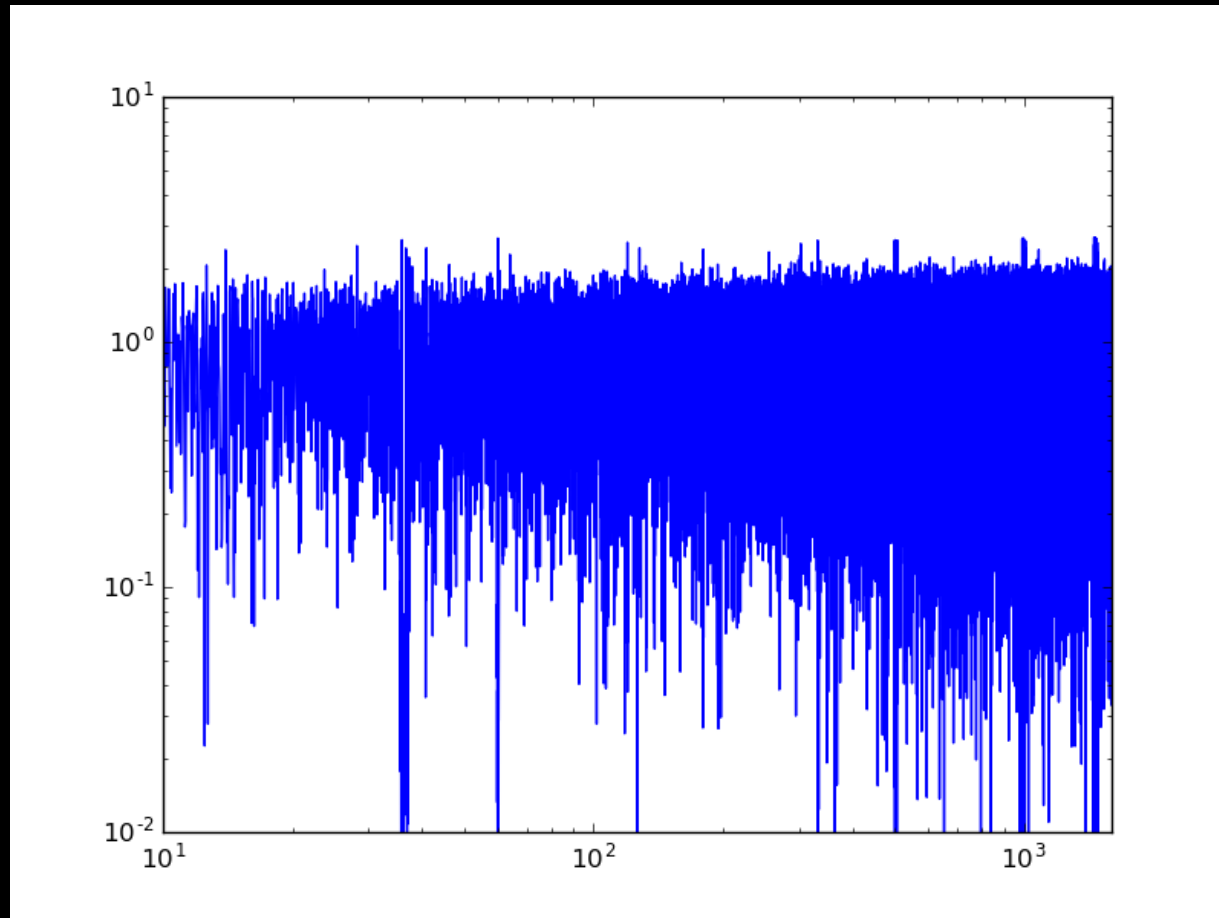
How Should We Estimate Noise?

- Windowing key to avoiding FFT ringing
- smooths out spectral features
- Noise large per mode in FT, so we have to average
- What are your thoughts on averaging?

Smoothing PS

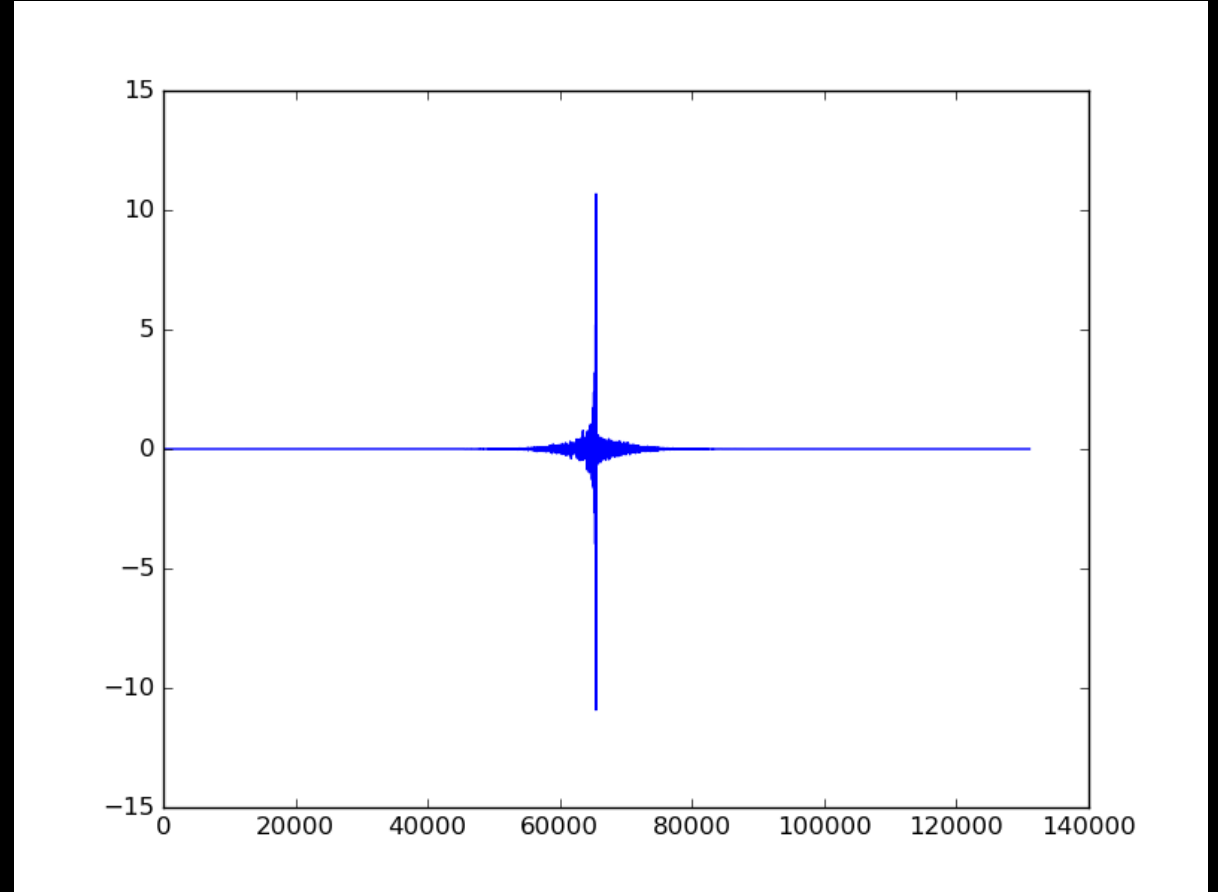
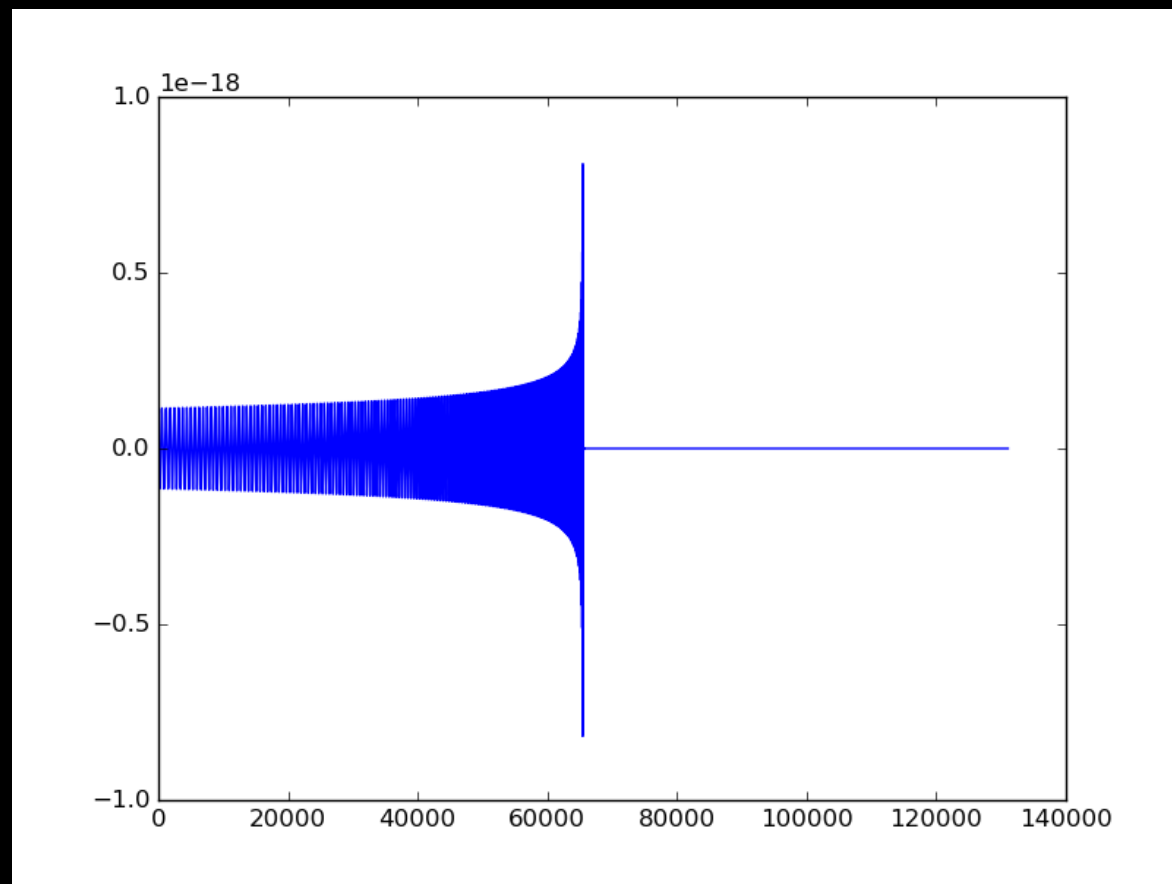
- Take $|FT|^2$, which is an estimate
- Smooth by convolving with an extended function.
- Thoughts on the function?

Pre-Whitened Data from Smoothed PS



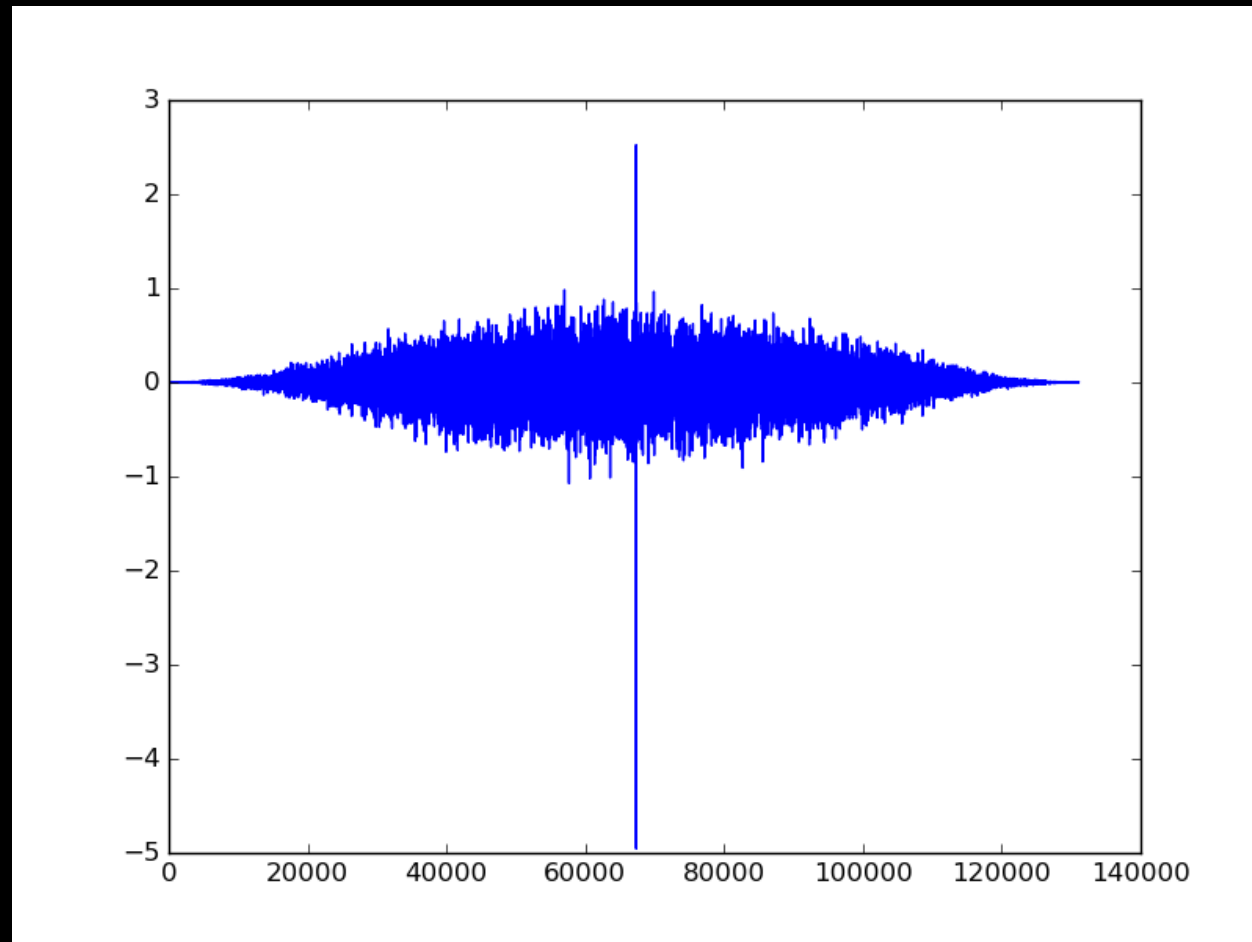
Left: Whitened FT of data. Looks not crazy. What are little nubbins sticking up?
Right: whitened data. Window shape is pretty obvious.

Pre-whitened template



- For this event, template is not small at start of data. Will this be a problem?
- Can look at pre-whitened version of template to get an idea.

Can Use for MF now



FFT Shift of matched filter output. We found a GW!

Averaging PS

- Break PS up into small chunks so we have many
- Take the FT of each chunk
- Add the FT²s together.
- How do we apply this (short) PS to original data?
- Qualitatively, how do we relate this PS estimate to smoothed one?

More Windowing

- Usual windows taper every sample.
- Reduces power in a way we probably aren't happy with
- How could we modify window to make this less of an issue?
- Let's try this on data...

Normalizations

- Properly normalizing noise can require care. I usually check with white noise.
- N^{-1} for white noise with $\sigma=1$ should be identity.
- Variance of FT is sum over data = $\sigma^2 N_{\text{data}}$. In Fourier space, we want $N(k) = \text{Var}(F(k)) / N_{\text{data}}$. Not just $\text{Var}(F(k))$
- Window function: $\text{Var}(F)$ for white = $\sum (\sigma^2 W^2)$.
- If you want the real-space variance to be correct where $W \sim 1$, you'll need to use $\sum (W^2)$ as your normalization.